🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.5

Explore Numerical Expressions with Exponents

Lesson 6-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.
Language Objective: I can explain my work using the words exponent, base, power, and evaluate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Powers and Exponents | **Exponent** | **Base** | **Power** | **Evaluate**

👆 *Tap any word to see what it means and a picture.*

1. A power writes repeated multiplication with a base and an ________.

2. The small raised number that shows how many times the base is multiplied is the ________.

3. The number being multiplied repeatedly in a power, like the 3 in 3^4, is the ________.

4. A number written with a base and an exponent, like 3^4, is a ________.

5. To find the value of an expression, like $2^3 = 8$, is to ________ it.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How large is the file after 5 layers? Why does the file size grow so quickly?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Powers and Exponents** — A power uses a base and an exponent to show repeated multiplication.
  Potencias y exponentes — Una potencia usa una base y un exponente para mostrar multiplicación repetida.

- [ ] **Exponent** — A small number that tells how many times to multiply a number by itself.
  Exponente — Un número pequeño que dice cuántas veces multiplicar un número por sí mismo.

- [ ] **Base** — The number that gets multiplied by itself.
  Base — El número que se multiplica por sí mismo.

- [ ] **Power** — A number written with a base and an exponent, like 2^3.
  Potencia — Un número escrito con una base y un exponente, como 2^3.

- [ ] **Evaluate** — To find the value by putting numbers in for the letters.
  Evaluar — Encontrar el valor poniendo números en lugar de las letras.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

After 5 layers the file is ____ MB because $4 \times 2^5 = 4 \times$ ____ = ____. The file grows quickly because each layer ____ the size.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The exponent counts how many times we multiply, not what we add.

Evidence: Students connect each switch flip to one more factor of 2 and explain that $2^3 = 2 \times 2 \times 2 = 8$, distinguishing repeated multiplication from repeated addition (which would give 6).

Reasoning: This evidence connects the definition of exponent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **After 5 layers the file is ____ MB because $4 \times 2^5 = 4 \times$ ____ = ____. The file grows quickly because each layer ____ the size.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Powers and Exponents
2. **Notes 2:** Exponent
3. **Notes 3:** Base
4. **Notes 4:** Power
5. **Notes 5:** Evaluate
6. **Watch for this mistake:** *A common mistake in Powers and Exponents is multiplying the base by the exponent instead of using the base as a repeated factor — for example, thinking* $2^3 = 2 \times 3 = 6$ *instead of* $2^3 = 2 \times 2 \times 2 = 8$*. Before you submit, rewrite the power as repeated multiplication first, then multiply step by step.*
7. **Try It 1:** A. 7^3 — *The base 7 is used as a factor 3 times, so the power is* 7^3*.* $7^3 = 7 \times 7 \times 7 = 343$*.*
8. **Try It 2:** A. 64 — $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$*. The exponent 6 counts how many times 2 is a factor.*